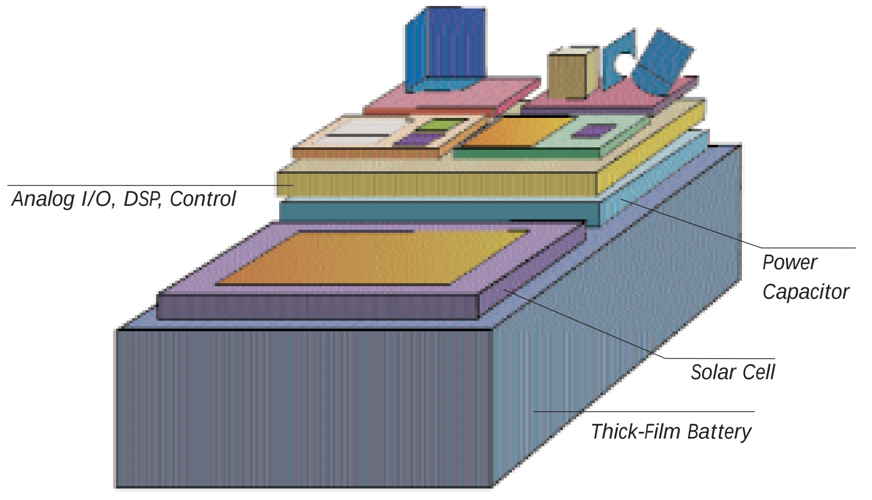

The Smart Dust project motes, being worked upon by the University of California, Berkeley, is going to be a self-contained, millimeter-scale sensing and communication platform for a massively distributed sensor network

Sands of time

Of course, the challenges are still there—current technologies restrict the size of the sensors. It would take some time to create dependable sensors that resemble household dust. Since sensors on the wireless networks (nodes), will need to operate autonomously, they need to have dependable power supplies. The Sensor Webs project for instance, does not look at microscopic devices, yet the implementation at Huntington Gardens was done with solar panels, and each sensor also had a battery.

Another crucial problem is that of actual networking. The sensors currently deployed on the Sensor Webs project, communicate using a mesh-type network, where each sensor links up to other neighbouring sensors on the local net via radio-based connections. With careful network management features built in, the wireless network is maintained as nodes get added or come off the network. Since it makes sense for each sensor to tune in, and talk, as and when required, the network will be dynamic in nature. To keep the network up and going, each sensor is being made smart enough to notice if a neighbouring sensor is going down, in which case, all the sensors around adjust their sampling rates in order to make up for the loss, whilst informing central command about it.

The data routing protocols will be customised for select applications, and will depend on the type of the sensor network. The communication protocols are still not standardised; indeed, there are several factors to be considered for the right protocol, such as battery power consumption, the amount of traffic, efficiency of transmission, as well as the numbers of nodes on the network. Since this is going to be a dynamic and relatively autonomous network, the protocols being considered range from the generalised Time Division Multiple Access (TDMA) and Code Division Multiple Access (CDMA) protocols, to the more exotic Ultra-Wideband (UWB), apart from the basic radio or cellphone-style communications. Eventually though, each set of applications will use a dedicated protocol.

Information will be managed conveniently on a localised small-scale network. The Sensor Webs project, for instance, can have all the results sent over to a central server, from where, if needs be, each individual sensor can be monitored over the Internet. However, if the network comprises huge numbers of dust-sized sensors scattered just about all over the world, the sheer volume of information is going to be huge. Anticipating this, teams at PARC are working on a prototype search engine that will allow one to query this vast network. It could be the sensor world's Google, and be just as simple to operate. The query posed will be translated into a language that nodes can understand. Intelligent networks will then send back a reply based on the information just learnt. For example, you could query the Internet using this search feature, and find out if you have enough vegetables in your cold storage, or a soldier could query and find out if adequate ammunition is available in the local stockyard.

Dust at work

There's a large market of applications waiting for Smart Dust, and researchers are scrambling to tailor make solutions for it. Funded by DARPA, Smart Dust may be used as battlefield sensors in applications of military importance, as may the Collaborative Sensing project at PARC. In both cases, battlefield sensor style devices seem to be war's new toys.

Other possibilities include improved Human-Computing interfaces owing to tiny sensors mounted under your nails. Sensor-enabled interfaces for the disabled will give them a level of freedom in communications, and in personal interactions with other people.

Inventory control could be a case of a sensor-in-box talking to a sensor-in-shelf, thus enabling you to query and receive accurate inventory checklists instantly. The houses we live in could become intelligent, thanks to sensors that will monitor and accordingly adjust the temperatures inside the house to ensure the perfect day, weather, light…the possibilities are many.

Such sensors could monitor product quality and safety systems in factories, industrial areas and places containing hazardous materials. Sensors embedded in concrete, for example, could tell you when the structure has reached a fatal stress point. The University of California at Los Angeles (UCLA) now uses an intelligent sensor network to monitor seismic (earthquake-related) activity in the UCLA area. Built into buildings and programmed to detect structural damage, they could act as early warning devices. Also, plans are afoot to use it to scrutinise stockpiles and flow of weapons of mass destruction.

This is the beginning

Recent advances in micro-electronics, power supplies, wireless communication and other esoteric fields are working towards wiring up the world, and the possibilities of sensors ringing the world is a not-so-distant reality—from academia to industry, everybody is looking at this turning into a reality within the next decade. And ten years from now, cheap pervasive computing is just what we are going to get, from nano-scale sensor networks located right under our fingernails!

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